

Rishab Shukla

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SUMMARY

AI engineer with a backend background, now working on **LLM** systems: **MCP** tool servers, **RAG** pipelines and agents wired into deterministic **Spring Boot** services that do the deciding. Java, Python, **Kafka**, **Kubernetes**.

EXPERIENCE

Software Engineer

Feb 2026 – Present

IDLMeals

Fort Worth, TX (Remote)

- Engineered **MCP servers** exposing **14** internal recipe and ingredient tools to AI agents over JSON-RPC 2.0, with allow-listed tool scopes and permission boundaries governing backend operations
- Built production **RAG** workflows in Python with embeddings and **vector search** over **38K recipes** and **9K ingredient records** (~410K embedded chunks), enabling semantic retrieval before LLM inference
- Designed **7** AI-assisted workflows bridging **tool calling** and RAG with deterministic Spring Boot services, enforcing **Pydantic** schema validation, restricted tool execution and fallback handling
- Optimized a data intensive retrieval layer using **JPQL projections**, indexed lookups and request driven field selection, reducing API payload size by **90%** and improving median response time from **780ms to 320ms**

Software Engineer Intern

Feb 2025 – May 2025

IpserLab

Fort Worth, TX

- Integrated an **LLM powered** feature into an existing application using Python inference services and **REST APIs**, handling input preprocessing, context assembly and model response parsing across **4** application workflows
- Implemented **prompt templates**, structured outputs and response validation, evaluating against a **180-case** input set and raising valid structured responses from **82% to 97%**
- Identified and fixed a mobile animation stutter on the production site by replacing position-based animation with GPU accelerated CSS transforms, raising frame rate from **24fps to a steady 60fps**
- Created **13** reusable React components integrating asynchronous APIs and accessible UI patterns, removing roughly **600 lines** of duplicated frontend logic across **5** application screens

Software Development Engineer

Jul 2022 – Aug 2023

Netzwerk.AI

Gulbarga, India

- Developed Python **NLP pipelines** to preprocess, clean and segment ~**340K** unstructured documents into **2.8M** searchable text chunks, supporting large scale semantic search across learning and application content
- Programmed semantic retrieval services by batching embedding generation at ~**1.2K chunks/min** into a **6GB** vector index, using **similarity search** and metadata filtering to retrieve relevant content at scale
- Integrated Python based AI retrieval services with Java and Spring Boot REST APIs across **17** production endpoints, implementing authentication, validation and backend access controls for application workflows
- Decomposed **3** high-traffic workflows into independently scalable services, offloading downstream processing through **Kafka** and horizontally scaled **Kubernetes** workers to isolate failures and increase peak throughput by **2.4 times**
- Established **CI/CD** pipelines with **Azure DevOps**, Docker and Azure, automating builds, testing, static/dependency/container scanning, artifact publishing and deployment; cut release time from **37 min to 9 min** and raised deployment frequency from **2 to 6 releases/week**

PROJECTS

RAG Chatbot | Python, Streamlit, LangChain, FAISS

- Scaled a **RAG** chatbot over a **1M** document corpus with batched ingestion, chunking, embedding and **FAISS** retrieval pipelines, grounding LLM responses through **top-k semantic search** and re-ranking

Coder Buddy | LangGraph, Groq, Python

- Orchestrated a **multi-agent** coding system in **LangGraph** and **Groq** where planner, code generation and validation agents share **structured state**, turning natural language requirements into reviewed, multi-file code

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, SQL

AI / ML: LLMs, RAG, MCP, NLP, Embeddings, Vector Search, Tool Calling, LangChain, LangGraph, FAISS, Groq

Frameworks & Systems: Spring Boot, Hibernate, React, Streamlit, REST APIs, JSON-RPC 2.0, Kafka

Cloud & Tools: Azure, Azure DevOps, Docker, Kubernetes, CI/CD, Linux, Git, PostgreSQL, MySQL

EDUCATION

Northeastern University

Boston, MA

Master of Science in Information Systems

Sep 2023 – Dec 2025

Visvesvaraya Technological University

Belgaum, India

Bachelor of Engineering in Electronics and Communication Engineering

Aug 2018 – Jun 2022